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21/7/26 Ex NO:01 : Basic network commands

1) Aim:
Study of various network commands used in linux and windows.

arp-a :
ARP - address resolution protocol. It will show the IP address of your computer along with the IP address and MAC address of your router.

hostname :
This is the simplest of all TCP/IP commands It simply displays the name of your computer.

ipconfig /all :
This command displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter in your system.

nbt stat-a:
This command helps solve problem with Net BIOS name resolution

netstat :
Displays variety of statistics about a computer's active TCP/IP connections. It is a command line tool for monitoring network connection both incoming and outgoing as well.

nslookup:

It is a tool used to perform DNS lookups in linux. It is used to display DNS details, such as IP address of a particular computer, the MX records for a domain or the NS servers of a domain or the NS server of a domain.

Pathping:

Pathping is unique to window's and is basically a combination of ping.

Ping:

This command is to test connectivity between two nodes.

Route:

Route command is used to show / manipulate the IP routing table. It is primarily used to setup static routes to specific host or network via an interface.

Linux commands.

IP:

The IP command is one of the basic commands every administrator will need in daily work, from setting up new system and assignment IPs to troubleshooting, manipulate routing, plus display network various devices interface and tunnels.

ifconfig:

The ifconfig command is a staple in many sysadmin's tools belt for configuring and trouble shooting networks. It has since been replace by the ip command discussed above.

mtr:

Matt's trace route is a program with a command line interface that serves as a network diagnostic and trouble string tool. This command combines the functionality of the ping and traceroute commands. You will get more information about route. If you see a sudden increase in response time or packet loss, than there is bad link somewhere

tcpdump:

tcpdump is a command designed for capturing and displaying packets.

Ping:

Ping is a tool that verifies IP level connea to another TCP/IP computer by sending Internet control message protocol (ICMP) Echo request message. The receipt of corresponding echo reply message is displayed along with round-trip times. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability and name resolution.

Ping google.com.

Result:

Thus the various network commands used in Linux and windows are studied and the output is verified.

Ex:NO:1 Study of network commands.

Windows commands:

1. arp-a : ARP is short term of address resolution protocol.

O/P: Interface : 10.16.56.50 ... 0x9

Internet address	physical address	Type
10.16.56.133	ec-12-17-17-f2-3c	dynamic
10.16.56.233	ff-ff-ff-ff-ff-ff	static

2. hostname : This is the simplest of all TCP/IP command. It will display the name of your computer.

O/P: LAPTOP-7003BK7N.

3. ipconfig/all : This command displays configuration information about your TCP/IP connection.

O/P:

Window IP configuration

Host Name LAPTOP-7003BK7N

Node type ..... Mixed

4. nbstat-a : This command helps solve problems with netbios name revolution.

O/P:

Ethernet 2:

Node ipaddress : [192.168.56.1] Scope Id : []

Host not found

5. netstat : Network statics

netstat display a variety of statistics about a computer active TCP/IP connection.

O/P: Active connection.

Proto	local address	Foreign address	state
TCP	10.16.56.50.49444	4.213.25.242:https	Established

6. nslookup : It is used to display the DNS detail. such as Ip address of a particular computer

O/P: Server : Unknown
Address: 10.16.56.133.

7. Pathping : Pathping is a unique to windows and its basically a combination of the ping and Tracert commands.

cmd : pathping -g google.com

O/P: Tracing route to google.com [142.250.66.14] over a maximum of 30hps.

Trace complete.

8. Ping : (Packet Internet groper) command is the but way to tat connectivity between two nodes.

Ping gogle.com

O/P: pingping

9. Route : route conamand is used to shoco/ manipulate the Ip routing table. It is primarily used to setup static route to specific network via an interface.

: manipulate network routing tables

Route [-f] [-p] [-4] [-6] command [destination]

Cmd: Route

13... 0A 00-27 00 00-0d ... virtual Box that only ethernet adapter.

Linux commands:

1. Ip commands

1. cmd: ip address show - displays all Ip addresses of network interfaces.

O/P: eth0: inet 192.168.1.10/24.

2. ip address add 192.168.1.254/24 dev enp0s3 - assign an ip address to an interfaces.

O/P: Ip address added successfully.

3. ip link set eth0 up - enables the network interface

O/P: eth0 state UP.

4. Ip link set eth0 down - disables the network interface.

O/P: eth0 state down.

5. ip link set eth0 promicon - Enables promiscuous mode.

O/P: promiscuous mode enabled.

6. Ip route add default via 192.168.1.254 dw eth0 - adds the default gateway.

7. ip route get 10.10.1.4 - shows the route to a destination Ip

O/P : 10.10.1.4 via 192.168.1.254 dev etho.

2. If config

displays network interface configuration.

O/P: etho link encap; Ethernet inet 192.168.1.10

3. mtr

1. mtr google.com - shows route statistics to host.

O/P : Host : localhost % avg Bet.

2. mtr -g google.com - shows numeric Ip address only

O/P: 142.250.183.286

3. mtr -c 10 google.com - shows both hostname and Ip address

O/P: google.com (142.250.183.206).

4. mtr -c 10 google.com - sends only 10 probe packets.

O/P : sent 10 packets.

3. tcpdump:

1. dnf install -y tcpdump - install tcpdump.

2. tcpdump -D - Cat available capture interface.

O/P : 1. etho 2. Io.

3. tcp dump -i etho -c 10 - capture packets to 1 from a host

O/P: 10 packets captured.

4. tcpdump -i etho src host 8.8.8.8 - captures packets from a host

O/P : 8.8.8.8 > 192.168.1.10.

5. tcpdump -i etho dst host 8.8.8.8 - captures packets to a host

O/P : 192.168.1.10 > 8.8.8.8.

6. tcpdump -i etho port 53 - captures DNS traffic.

O/P : DNS captured

7. tcpdump -i etho -c 10 host www.google.com and port 443 - captures HTTPS packets

O/P : TCP 443 captured.

8. tcpdump -i etho port not 53 and not 25 - captures all traffic except DNS and SMTP.

O/P : TCP/IP packets displayed.

5. Ping :

1. ping google.com - checks connectivity to a host

o/P : 64 bytes from google.com

2. ping -c 10 google.com - sends only to ping requests.

O/P : 10 packets transmitted - 10 recorded

nmcli :

1. nmcli connection show - Lists all network connections.

O/P : wired connection 1.

2. nmcli connection modify "wired connection 1" modifies an exciting connection.

3. nmcli connection show - displays connection details.
O/P: ipv4. method auto.

4. nmcli connection modify "wired connection 1" ipv4. method auto - configures DHCP.
O/P: ipv4 method set to auto

5. nmcli connection modify "wired connection 1" ipv4. method manual ipv4. address 192.0.2.1/24 ipv4 gateway 192.0.2.254 ipv4. dns 192.0.2.200 - configures a static Ipv4 address
O/P: Ipv4 settings updated.

6. nmcli connection modify "wired connection 1" ipv6. method auto - configures automatic Ipv6.
O/P: Ipv6 method set to auto.

7. nmcli connection up Internal_LAN - Active the connection profile.

Result:
The above both the windows and linux commands are executed and outputs are completed successfully.

28/11

17/07/26

Ex.No.02 Study of different types of Network cables

Aim:

To study different types of Network cable.

a) Understand different types of network cables.

1. Unshielded Twisted Pair (UTP) cable.
2. Shielded Twisted Pair (STP) cable.
3. Coaxial cable
4. Fibre optic cable.

Cable Type	category	Max Data Transmission	Advantage/ Disadvantage	Application / use.
UTP	category 3	10bps	Advantage: Cheaper in cost Easy to install as they have a smaller overall diameter. Disadvantage: More prone to (EMI) Electromagnet interference and noise	*10 Base T-ethernet *Fast ethernet gigabit ethernet *Fast ethernet gigabit ethernet.
	category 5	upto 100 Mbps		
	category 5e	1 Gbps		
STP	category 10 6, 6a	10Gbps	Advantage: * shielded. * Faster than UTP * less susceptible to noise and interference	* Gigabit ethernet 10G ethernet (55m) widely used in data centres.

SSTP	Category 7	10 Gbps	Disadvantage: * Expensive * Greater installation effort	* Gigabit ethernet, 10G Ethernet (100 m)
Coaxial Cable:	RG-6 RG-59 RG-11	10-100 mbps	Advantage: * High bandwith * Immune to interference * Low loss bandwidth * versatile. Disadvantage: * Limited distance * cost * size in bulky	* speed of signal is 500m. * Television network * High speed internet connection.
Fibre optics Cable.	Single mode, multi mode	100 Gbps	Advantage: * High speed * High bandwidth. * High security. * long distance. Disadvantage: * Expensive. * Required skilled installer.	* Maximum distance of fibre optics cable is around 100 metre.

Result:

Thus, the study of different types of network cable studied successfully.

3/8/22

EX.NO: 03 Cisco Packet Tracer.

Aim:

To study the cisco packet Tracer simulation tool, understand its user Interface, design a simple network and analyze the behaviour of Hub, Switch and peer-to-peer.

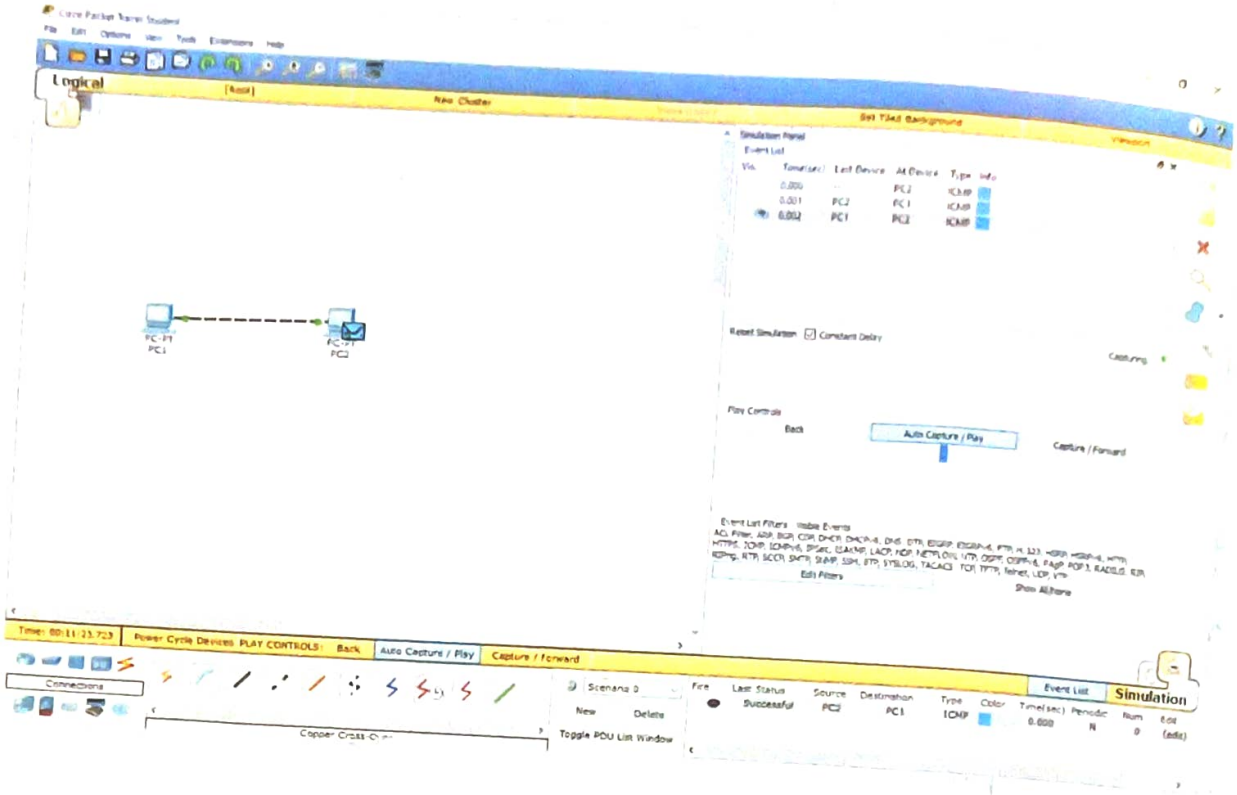
Cisco Packet Tracer:

Cisco Packet tracer is a network simulation and modeling tool. It allows us to create and test computer networks without using real networking equipment we can add PCs, switches, hubs, routers and other devices and observe how packets travel between them

①Peer-to-Peer:

Two computers communicate directly.

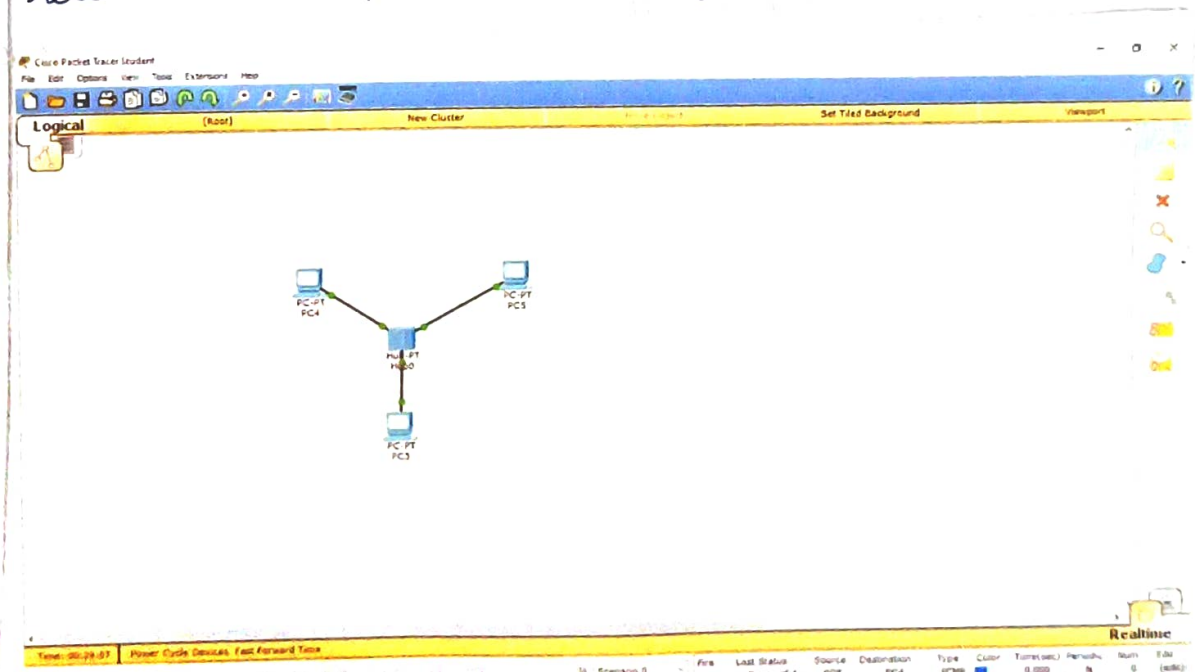
A peer to peer network was created by connecting 2 PCs directly. IP addresses were assigned to both PCs in the same network. The connectivity between the the 2 PCs was tested using ping command. successful replies were received, confirming That the two PCs could communicate directly with each other.



(2) Network using hub:

Connected multiple computers & forwards received data to all ports.

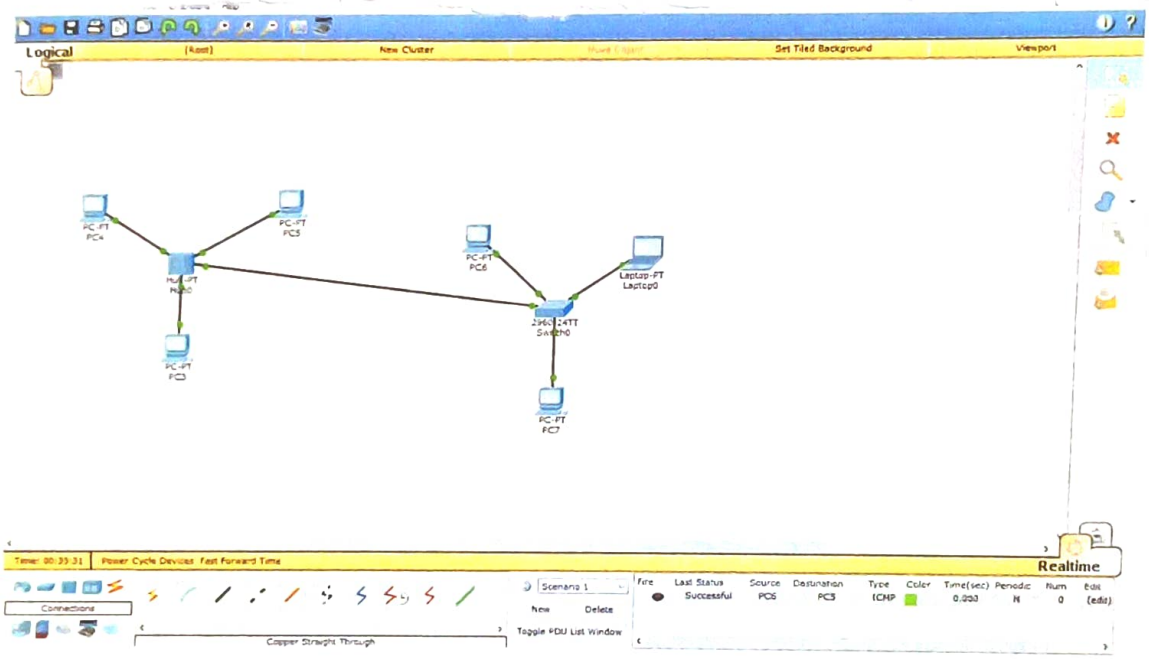
A network was created by connecting 3 PCs to a Hub using copper straight through cables. IP addresses were assigned to all PCs. When data was sent from one PC to another the Hub forwarded the packet to all connected ports. The destination PC received the packet successfully.



(3) Networking Switch:
Connects computers and forwards data forward the required destination.

A network was created by connecting 3 PCs to a switch using copper straight through cables. IP addresses were assigned to all PCs and connectivity was tested using ping. The switch forwarded the packet to appropriate destination.

(4). Hub and Switch:
Both hub and switch were used in the network - the packet forwarding behaviour of both devices was observed and compared.



Result:
The peer to peer, Hub & Switch networks were successfully designed and their communication behaviour was observed using cisco packet Tracer.

9/8/21

Ex.No:04 Setup and configure a LAN cloud area Networks using a switch and ethernet cables in your lo

Aim:

To ~~steep~~ setup and configure a LAN (local area network) using a switch and ethernet cables in cisco packet tracer and verify the connecting between the devices.

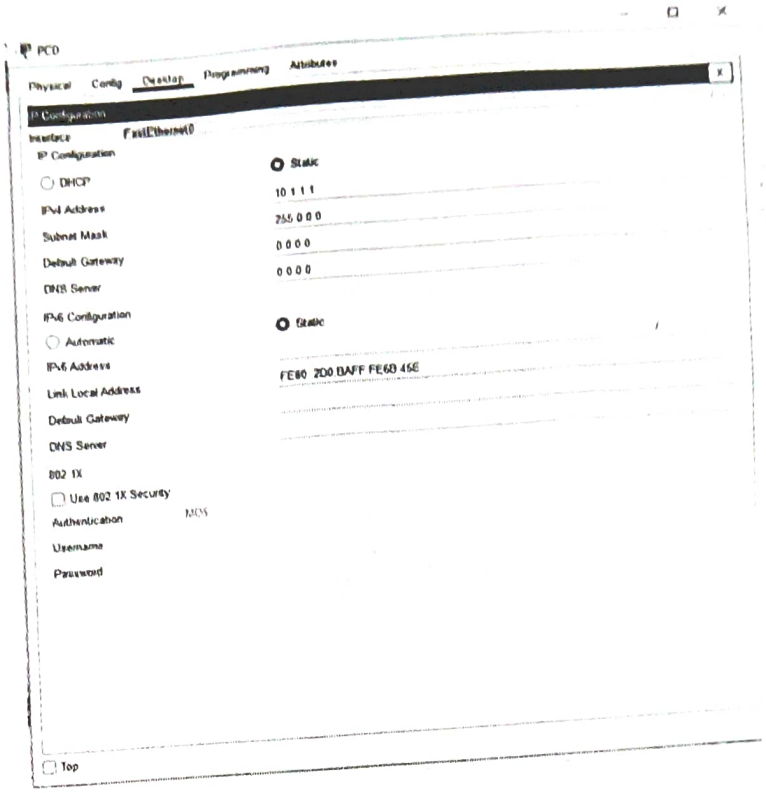
Procedure:

Step 1: Open cisco packet tracer and create a LAN topology using 4 PCs and 1 switch.

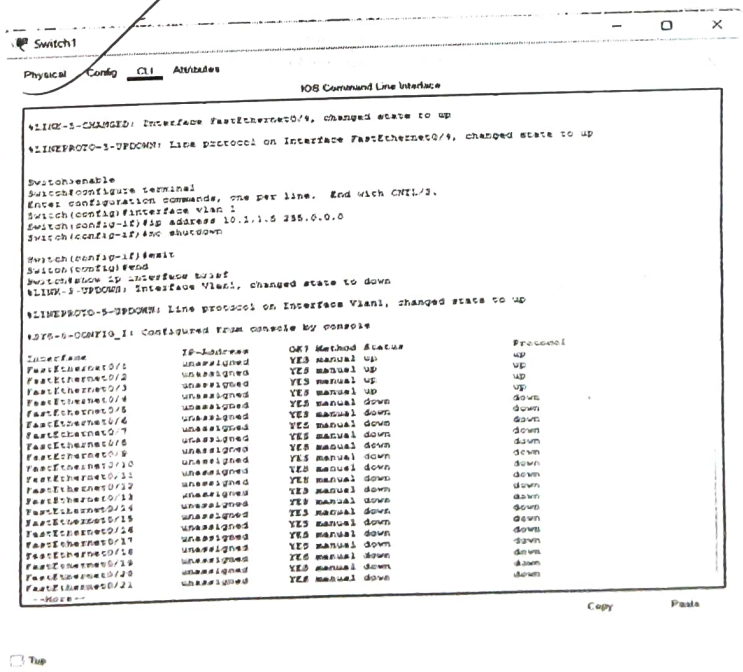
Step 2: Connect all the PCs to the switch using copper straight-through ethernet cable.

Step 3: configure the IP address for each PC using the following settings.

Device	IP address	subnet Mask.
PC0	10.1.1.1	255.0.0.0
PC1	10.1.1.2	255.0.0.0
PC2	10.1.1.3	255.0.0.0
PC3	10.1.1.4	255.0.0.0

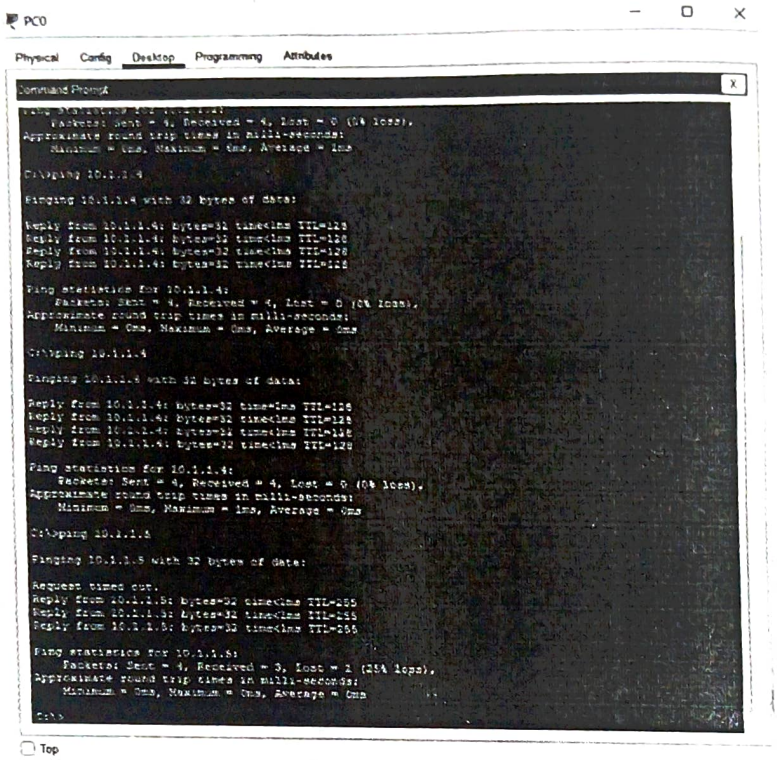


step4: configure the switch using the CLI
Assign the switch 1 interface the IP address
10.1.1.5 with subnet masks 255.0.0.0 and
enable the interface using the no shutdown command



Step 5: Verify the connectivity between the PCs and switch using the ping command from the command prompt.

Ex: Ping 10.1.1.4
Ping 10.1.1.5



Result:

Thus, a LAN successfully configured using a switch and ethernet cables in Cisco packet tracer and the connectivity between the devices successfully verified using Ping command.

10/8/22

EX NO:5 Packet capture tool: Wireshark.

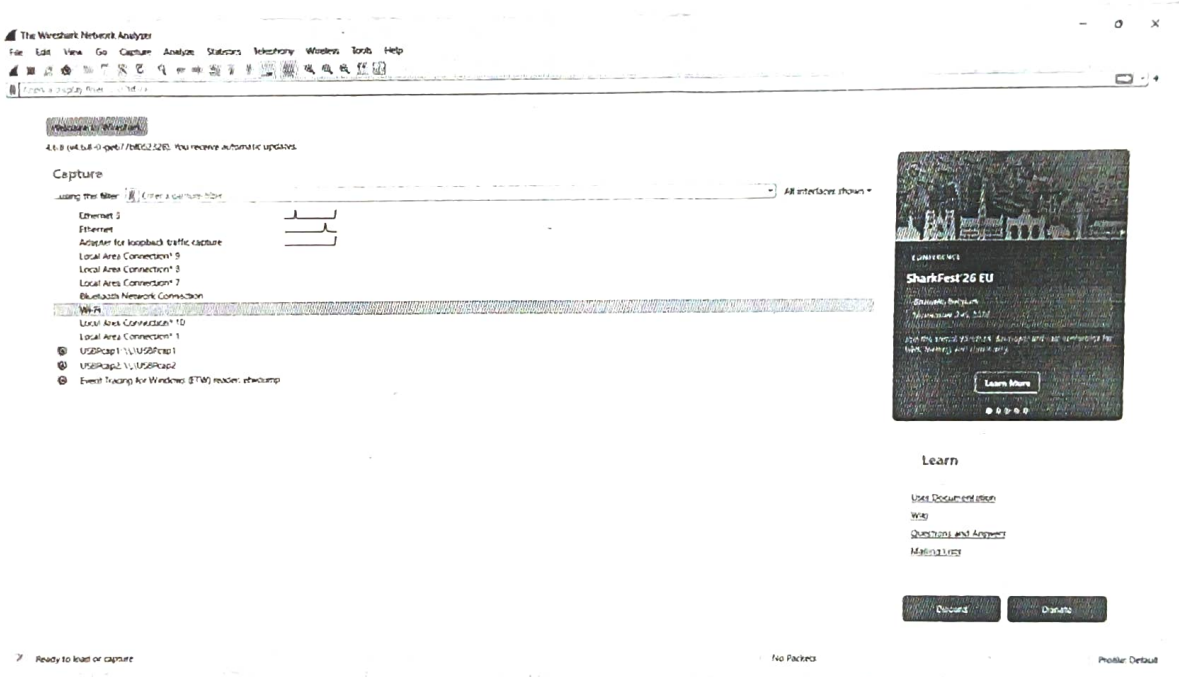
Aim:

To understand the features of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol stack.

Procedure:

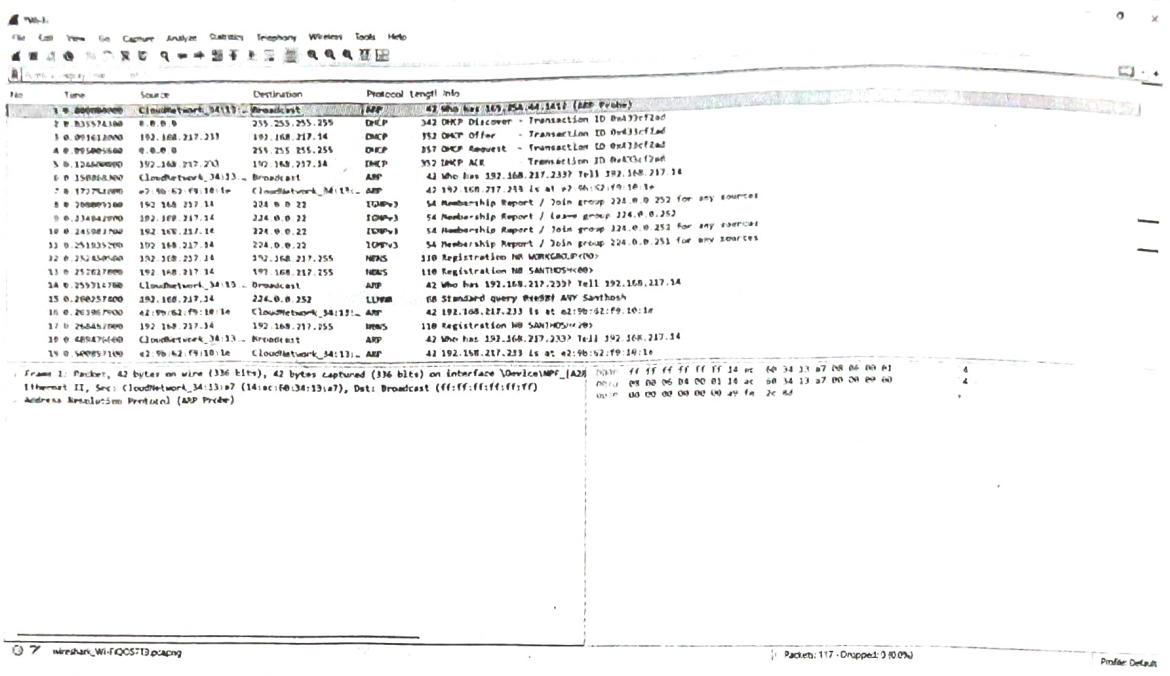
Step 1: Open wireshark and select the wifi network interface.

Step 2: Start the packet capture by double clicking the wifi interface.



Step 3: Observe the packets being captured in the wireshark window.

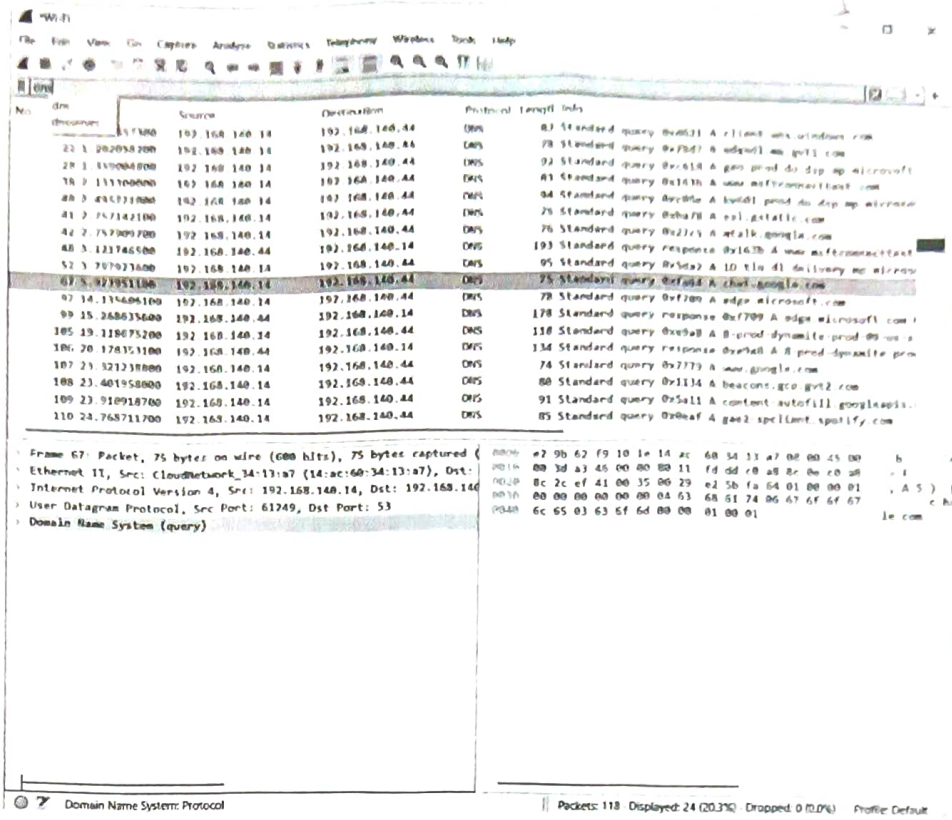
Step 4: Different packets such as ARP, DHCP, IGMPv3, NBNS and LLVVR can be observed in the packet list.



Step 5: Select any packet to view its detailed information such as frame, ethernet, IP, UDP/TCP and protocol details.

Step 6: Apply the display bar to display only DNS packets.

Step 7: Observe DNS query and packets The source and destination IP address, protocol, packet length and DNS information can be analyzed



Step 8:

Stop the packet capture after collecting the required packets.

Result:

Thus, networks packets work successfully captured and analysed using wireshark, and DNS packets over indentified using the display filter.

Ex:6 Error correction at data link layer

Aim:

Write a program to implement error detection and correction using Hamming code concept. make a bat run to input stream and verify error correction feature

Code:

```
data = input("Enter 4 data bits : ")
d1, d2, d3, d4 = map(int, data)
  P1 = d1 ^ d2 ^ d4
  P2 = d1 ^ d3 ^ d4
  P4 = d2 ^ d3 ^ d4
code = [P1, P2, P1, P4, d2, d3, d4]
Print("Hamming cod : ", .join(map(str, code))
rCved = set(map(int, input("Enter second code : ")))
S1 = rcved[0] ^ received[2] ^ reved[4] ^ reved[6]
S2 = reved[1] ^ reved[2] ^ recved[5] ^ Recved[6]
S4 = received[3] ^ received[4] ^ received[5] ^ received[6]
error = S4 * 4 + S2 * 2 + S1
if error = 0:
    Print("NO error")
else:
    print("Error at position :", error)
    received[error - 1] ^ = 1
    Print(corrected code : ", ".join(map(str,
received)))
```

o/p:

Enter 4 data bits 1001

Hamming code : 0011001

Enter received code : 1001011

Error at position : 4

Corrected code : 1000011

Result:

Thus, the hamming code was successfully generated, the error was detected and the bit was correctly placed.

17/8/23

Ex: 1 Flow control at data link layer

Aim:

Write a program to implement flow control at data link layer using sliding window protocol, simulated the flow of frames from one node to another.

Code:

Sender.py

```
message = input ("Enter message : ")
window = int (input ("Enter window size: "))
i = 0
while i < len (message):
    print ("\n Sender Buffer :")
for j in range (i, min (i + window, len (message))):
    Print ("Frame", j+1, ":", message [j])
    ack = int(input ("Enter ACK received : "))
if ack >= i+1 and ack <= min(i+ window, len (message)
    Print ("ACK accepted ")
    i = ack
else :
    Print ("Wrong ACK - Resending frame")
Print ("\n All frames sent successfully")
```

Receiver.py :

```
frame = int(input("Enter received frame number : "))
data = input("Enter frame data :")
expected = int(input("Enter expected frame number :"))
print("\n Receiver :")
if frame == expected :
  Print (" Frame", frame, "received")
  Print ("Data:", data)
  Print ("ACK", frame, "sent")
else :
    print ("Wrong frame")
    print ("NACK sent")
```

O/P:

Sender :

Enter message : hello

Enter window size : 3

Sender Buffer :

Frame 1 : h

Frame 2 : e

Frame 3 : l

Enter ACK received : 3

ACK accepted

Sender Buffer :

frame 4 : l

frame 5 : o

All frames sent successfully

Receiver:

Enter received frame no: 1

Enter frame data: h

Enter expected frame no: 1

Receiver:

Frame 1 received

Data: h

ACK 1 sent.

Result:

Thus the sliding window protocol was successfully implemented.

Ex: 8 IMAP to Discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the tryhackme platform

Aim:

To discover live hosts using Nmap scans (ARP, TCP/UDP) on the tryhackme platform.

Procedure:

1. Setup and environment Initiation:

Log in to the tryhackme platform and navigate to the target room: Tryhackme Nmap room.

Deploy the target virtual machine and establish a VPN connection using open VPN. or launch the browser based attachbox.

2. ARP scan for local host discovery:

Perform an ARP request scan to identify active hosts on the local network segments.

Command:

Sudo nmap PR -sn <target: IP(Subnet)>

3. ICMP. ping scan:

Send ICMP Echo request packets to check for TCMP Echo reply from active hosts.

Command:

nmap - PE - Sn <target - IP/Subnet>

4. ICMP timestamp ping scan:

Send ICMP Timestamp request packets to discover host that block standard ICMP Eho request.

Cmd:

Sudo nmap -PP -sn <target_IP/Subnet>

5. ICMP address ping scan:

Send ICMP address request packet to identify live systems by requesting subnet mask information.

Cmd:

Sudo nmap -PM -sn <target_ICP/Subnet)

6. TCP SYN Ping scan:

Send TCP SYN packets (typically to port) to discover systems based on SYN-ACK (or) RST responses.

Cmd:

Sudo nmap -PS -sn <taeget - IP/Subnet>

7. TCP Ack ping scan:

Send TCP Ack packets to elicit RST responses from active hosts, bypassing rules.

Cmd:

Sudo nmap -PA -sn <target_IP/subnet>

8. UDP ping scan:

Send empty UDP packets to destination ports to detect host via ICMP port unreachable response.

and:

sudo nmap -PV -Ah <target_IP/subnet>

9. Disable port scanning (Host discovery only)

We the sn flag with nmap to perform host discovery exclusively without triggering subsequent port scans.

And:

sudo nmap -sn <target_IP/subnet>

Result:

Thus, live ~~host~~ discovery using nmap scanning techniques (ARP, ICMP and timestamp, TCP SYN/ACK, and UDP ping scans) was successfully performed on the tryhackme platform.

24/8/26

Ex:9 Implementation of subnetting in cisco packet tracer simulator.

Aim:

To design, configure and simulate a subnetted network topology in cisco packet assign static IP addresses using IP subnetting multiple subnets and observe packet transmission across the subnet with and without a router.

Procedure:

1. Design Topology: open cisco packet tracer and place routers, switches and PCs on the workspace
2. Connect cables: Connect the PCs to switches and switches to routers interface using straight through cables.
3. Assign IP addresses to PCs: Go to desktop IP configuration each PC and assign

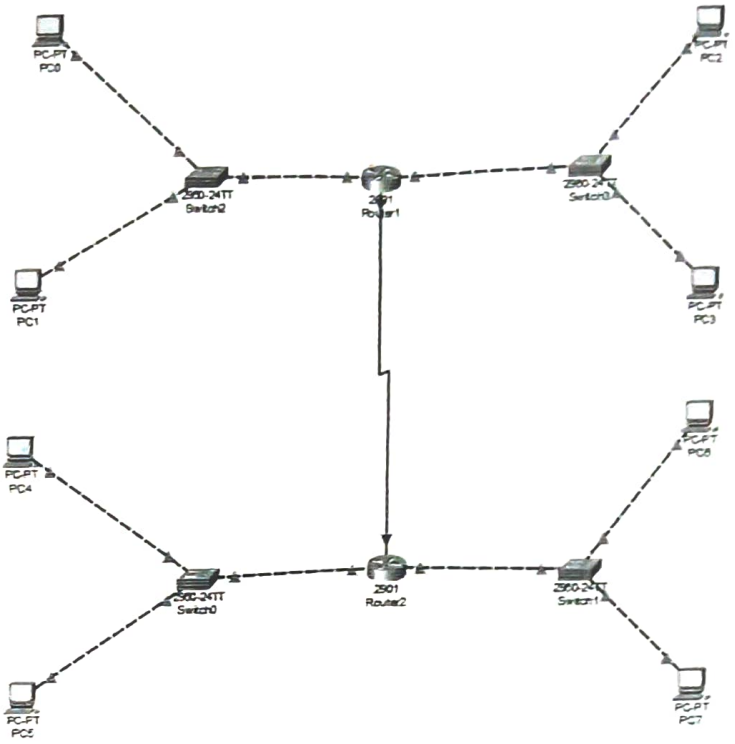
Subnet mask: 255.255.255.224 (/27)

IPs and gateways:

Subnet 1: PC IPs = 192.168.1.11 to 192.168.1.15 | Default gateway = 192.168.1.1

Subnet 2: PC IPs = 192.168.2.11, 1 192.168.2.15 (Default gateway = 192.168.2.1

4. Configure router : Open router CLI and configure interface.



5. Test communication:

With router: ping a pc in subnet 2 from subnet 1 using command prompt- (ping 192.168.2.11) ping will be successful.

Without router : disconnect the router and ping again. Ping will fail due to lack of inter-subnet routing.

Result:

The subnetted network topology was successfully designed and configured in cisco packet tracer.

25/8/25

Ex:12 A) Implement echo client server using TCP/UDP socket

Aim:

To write the program to implement client server using TCP/UDP Sockets

code:

server.py:

```
import socket
S= socket.socket()
S.bind(('127.0.0.1', 65432))
print("server ready...")
conn, addr = S.accept()
msg = conn.recv(1024)
print("Received :" msg.decode())
conn.send(msg)
conn.close()
  S.close()
```

client.py

```
import socket
S= socket.socket()
S.connect(('127.0.0.1', 65432))
S.send(b"Hello server")
reply = S.recv(1024)
print("Echo from server", reply.decode())
S.close()
```

Result

Thus, the above program is executed successfully.

31/8/21

B) Implement chat client server using TCP/UDP socket

Aim:

To write the program to implement the chat client server using TCP/UDP sockets.

code:

chat_server.py:

```
import socket
S = socket.select()
S.blind(('127.0.0.1', 65432))
S.listen()
Print("Waiting for client...")
Conn, addr = S.accept()
While True:
    msg = Conn.recv(1024).decode()
    if msg.lower() == 'bye':
        break;
    Print("client:", msg)
    reply = input("server:")
    Conn.send(reply.encode())
    if reply.lower() == 'bye':
        break

Conn.close()
S.close()
```

chat. client py.

```
import socket
s= socket. locket()
s. connect(('127.0.0.1', 65432))
while True:
    msg = input("client :")
    s-send(msg. encode())
    if. msg. lower() = 'bye':
        break.
    print("server :", reply)
    s-close()
```

Result

Thus the above program is executed successfully.

31/8/26

Ex: 13. Implement your own ping program

Aim

To write the program to implement your own ping program.

code:

Server.py.

```
import socket
server = socket.socket(socket.AF_INF, socket.SOCK_DGRAM)
server.bind(("127.0.0.1", 12000))
print("server is listening...")
while True:
    data, address = server.recvfrom(1024)
    print("Received:", data.decode())
    server.sending(data, address)
```

client.py.

```
import socket
import time
client = socket.socket(socket.AR_INET, Socket.Sock.awaf)
client.settimeout(1)
```

```
for i in range (1,6):
    start = time.time()
    client.sendto (f"Ping {i}", encode(),
("127.0.0.1", 12000))

    try:
        data, _ = client.recv(1024)
        rtt = (time.time() - start) * 1000
        print (f"reply: {data}, decode {i} / Time =
{rtt:.2f} ms")
    except socket.timeout:
        print ("Request timed out")

    client.close().
```

Result:

The above code is executed successfully.

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Ex:14 Write a code using RAW sockets to implement packet sniffing

Aim:

To write a code using RAW socket to implement packet sniffing.

Code:

Sniffer.py-

```
import socket
import struct
that = socket.gethostbyname(socket.gethostname())
sock = socket.socket(socket.AF_INET, socket.
sock.RAW, socket.IP)
sock.bind((HOST, 0))
sock.setsockopt(socket.IPPROTO_IP, socket,
IP_HDR, 1)
sock.ioctl(socket.SIO_RCVALL, socket)
print("Packet sniffing start...")
print("Host IP:", HOST)
print("Press ctrl+C to stop\n")
try:
    while True:
        packet, address = sock.recvfrom(65535)
```

```
ip_header = packet [:20]
iph = struct.unpack ("!BBHHHBBH4s4s", ip_head)
source_ip = socket.inet_ntoa (iph[8])
destination_ip = socket_inet_ntoa (iph[9])
protocol = iph[6]
print ("....")
print ("Source IP :", source_ip)
print ("Destination IP :", destination_ip)
Print ("Protocol :", protocol)
print ("Packet size :", len (packet))
print ("....")

Except Keyboard Interrupt:
    print ("\n sniffing shopped")

finally:
    lock.IOCH (socket. SIO_RCVALL, socket. RCVALLOFF)
    sock. close()
```

Result:

The above code is executed succesfully.

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Ex:10 Internetworking with Routers.

Aim:

To design and configure an internetwork using a router in cisco packet Tracer.

procedure:

Place one router and 2 PC's.

Connect both PC's to router using straight cables.

Assign IP addresses and subnet masks to router interface and PC's.

Configure the default gateway for each PC

Enable router interface using no shutdown

Test connectivity using ping command.

Route:

```
test
enable
configure terminal
interface Fast Ethernet 0/0
ip address 192.168.10.1
no shutdown
exit.
```

Observation:

The router connects 2 different network and transfer packets

Result:

Thus the experiment was successfully executed.

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Ex.11- Routing Information Protocol.

Aim:

To simulate a static routing configuration using cisco packet tracer.

Procedure:

Open cisco packet tracer and create the given network topology.

Connect 3 routers and PC's using suitable cable.

Configure IP addresses for routers and PC's

Enable all routers interface using no shutdown.

Configure static routes on router1 and router 2 using IP mobile.

Set main and backup route using different administrative distance value.

Test connectivity.

Use tracer to verify routing packets

Result:

Thus the experiment was carried out successfully.

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